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Piotr Dybka

SGH Warsaw School of
Economics, Poland

Stanisław Bartha

EY Economic Analysis Team

Maciej Łopusiński

SGH Warsaw School of
Economics, Poland

Radosław Kotkowski

Institute of Economics,
Polish Academy of
Sciences, Poland

Substitution or market creation? Causal evidence from Poland's domestic mobile payment scheme





Piotr Dybka Stanisław Barthä Maciej Łopusiński Radosław Kotkowski

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Keywords: Mobile Payments, Synthetic Control Method, Financial Innovation, Technology Diffusion, Market Creation, Payment Sovereignty

JEL Codes: G21, O33, E42



Substitution or market creation? Causal evidence from Poland's domestic mobile payment scheme

Piotr DYBKA^{1,*} , Stanisław BARTHA², Maciej ŁOPUSIŃSKI¹ , and
Radosław KOTKOWSKI³ 

¹SGH Warsaw School of Economics, Poland

²EY Economic Analysis Team

³Institute of Economics, Polish Academy of Sciences, Poland

*Corresponding author (pdybka@sgh.waw.pl)

Abstract

Do mobile payment innovations merely shift transactions from cards to mobile interfaces, or do they expand the frontier of digital commerce by converting previously cash-only exchanges? We address this question by exploiting the introduction of BLIK, that is a domestic mobile payment scheme in Poland, as a quasi-natural experiment. Using the penalized synthetic control method, we construct a counterfactual based on a weighted donor pool of 21 European economies to estimate the aggregate causal effect of the system over the period 2015-2024. Our two-stage identification strategy separates the technology effect from a concurrent supply-side intervention: the Cashless Payment Support Program (CPSP), which subsidized point-of-sale terminal deployment from 2018 onward. The synthetic counterfactual tracks Poland closely in the pre-treatment period (2000-2014), diverging progressively after 2015. By 2024, the aggregate gap reaches approximately 8.9 percentage points of Household Final Consumption Expenditure. Decomposing this gap at the 2023 endpoint, supply-side infrastructure subsidies account for approximately 2.7 percentage points (approximately 47%), while the net technology effect of the mobile payment standard amounts to approximately 3.1 percentage points (approximately 53%) of Household Final Consumption Expenditure. This net effect is consistent with genuine market creation, rather than mere substitution of existing card flows. The results suggest that domestic mobile payment schemes can serve as effective instruments for financial digitization, with implications for the design of sovereign payment architectures across Europe and emerging economies.

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1. Introduction

The migration toward digital retail payments marks a fundamental structural shift in the modern financial landscape (Rogoff, 2016). While the decline of cash is a global phenomenon, the trajectory of this decline varies significantly across economies (Kotkowski et al., 2025), shaped by a range of factors including payment card acceptance infrastructure, consumer preferences and demographics, financial incentives, and institutional characteristics (Bagnall et al., 2016). One factor that may be contributing to this divergence is the emergence of domestic mobile payment schemes, which have increasingly challenged the dominance of global card networks (e.g., Visa, Mastercard). Yet the economic impact of these innovations remains debated. A central ambiguity in the literature is whether they act primarily as substitutes, i.e. merely shifting volume from existing card infrastructure to mobile interfaces, or as mechanisms of market creation, converting previously cash-only transactions into digital flows in segments where card payments were impractical (Agarwal et al., 2019).

The scale of potential impact is illustrated by sovereign instant payment infrastructures operated or mandated by central authorities. Brazil’s Pix, launched by the Central Bank of Brazil in November 2020, reached 114 million individual users within 15 months and brought approximately 50 million previously cash-only consumers into the digital payment ecosystem, which suggests substantial market creation alongside the substitution of existing card and bank transfer volumes (Duarte et al., 2022). By 2024, the system processed over 63 billion transactions annually, and as of mid-2025 it had surpassed 175 million registered users, i.e. approximately 93% of Brazil’s adult population (Central Bank of Brazil, 2025). India’s Unified Payments Interface (UPI), operated by the National Payments Corporation of India since 2016, has grown from fewer than 3 million transactions in its first year to approximately 209 billion in calendar year 2025, making it the largest real-time payment system in the world (National Payments Corporation of India, 2026). Górka (2025) provides a comparative overview of the critical success factors behind these and other instant payment systems, identifying regulatory support, open ecosystems, and network effects on both the supply and demand sides of the market as key drivers of rapid adoption. The sheer scale of Pix and UPI demonstrates the policy stakes involved in sovereign payment architecture. However, both systems were centrally mandated and operate in institutional contexts, including zero-fee public utility models, mandatory bank participation, and large unbanked populations, that limit the generalizability of their experience to market-driven schemes in advanced economies.

Alongside these centrally operated infrastructures, a parallel development has unfolded in Europe: the proliferation of domestic mobile payment schemes, typically governed by bank consortia and operating on account-to-account (A2A) rails that bypass global card networks. As documented in Section 3, at least 19 such schemes are currently operational

across European economies, of which 13 are operated by bank consortia (Tables 1 and 2). A further 22 schemes have been discontinued, largely owing to insufficient network adoption or subsequent consolidation. Notably, all telecommunications- and corporate-led ventures have collapsed, suggesting that commercial sustainability depends critically on bank participation and the associated trust infrastructure. The ongoing consolidation wave, including GiroPay’s closure in Germany, Paylib’s replacement by Wero in France, and the Vipps–MobilePay merger in Scandinavia-, reflects both competitive pressure from pan-European initiatives and a broader trend toward larger, multi-country platforms. The European Payments Initiative’s Wero wallet now seeks to federate several of these national schemes into a continental standard (European Payments Initiative, 2025). The breadth and persistence of these efforts (spanning virtually every major European economy) signals both the strategic importance attached to payment sovereignty and the absence of settled evidence on the economic returns to investment in domestic payment infrastructure. Despite the proliferation of bank-consortium schemes across Europe, no study to date has provided causal evidence on whether any individual scheme has expanded digital payment volume beyond what would have occurred through organic card adoption alone.

This paper addresses this gap by examining the introduction of BLIK, a Polish mobile payment scheme launched in 2015, as a natural experiment in payment system innovation. BLIK is a domestic payment scheme and system operated by Polski Standard Płatności (PSP), a consortium of six major commercial banks. Unlike BigTech wallets (e.g., Apple Pay, Google Pay), which tokenize existing card credentials and operate as overlays on global card rails, BLIK functions as a sovereign domestic standard: users generate a temporary six-digit code within their mobile banking application, which is then used at point-of-sale terminals, ATMs, or e-commerce checkouts. The system’s clearing and settlement architecture varies by transaction type: peer-to-peer transfers are processed in real time through Express Elixir, Poland’s instant payment system operated by KIR (Górka, 2025), while contactless payments at the point of sale were introduced in July 2021 and operate in cooperation with Mastercard. By end-2024, BLIK had reached 18.5 million active users and accounted for approximately 67% of domestic online transaction value, while its share of cashless point-of-sale transactions is still modest at 6.3%, but it has been growing rapidly since the introduction of the contactless feature (Van Hove et al., 2026). This combination of rapid diffusion, institutional distinctiveness, and architectural independence from global card networks makes Poland a valuable case for isolating the causal effects of a sovereign payment standard on the volume and value of digital transactions.

Our empirical strategy employs the penalized synthetic control method (SCM) to estimate the causal effect of BLIK’s introduction on the total value of electronic transactions as a share of household consumption. Following the regularization framework proposed by Abadie (2021), we construct a “Synthetic Poland” from a donor pool of 21 European economies that did not experience a comparable domestic mobile payment shock during

the sample period (2000-2024). A two-stage identification strategy then separates the technology effect from the concurrent Cashless Payment Support Program (CPSP), which subsidized point-of-sale terminal deployment from 2018 onward. By 2024, the aggregate gap between Poland and its synthetic counterpart reaches approximately 8.9 percentage points of Household Final Consumption Expenditure (8.6 pp bias-corrected). Decomposing this gap at the 2023 endpoint, the CPSP infrastructure subsidies account for approximately 2.7 percentage points (approximately 47%), while the net technology effect of the mobile payment standard amounts to approximately 3.1 percentage points (approximately 53%). One important caveat frames these estimates: because the Polish outcome variable captures both card and BLIK transactions while donor countries are measured through card payments only, the estimated gap represents an upper bound of the pure net technology effect. Nevertheless, the net effect is consistent with genuine market creation. The digitization of transactions that would likely have remained in cash absent the innovation. The results suggest that domestic mobile payment schemes can serve as effective instruments for financial digitization, with implications for the design of sovereign payment architectures across Europe and emerging economies.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 details the institutional background. Section 4 describes the data and donor pool. Section 5 outlines the econometric framework. Section 6 presents the results, and Section 7 discusses the mechanisms. Section 8 outlines limitations, and Section 9 concludes.

2. Literature Review

2.1. Real Economic Effects of Mobile Payments

While early literature viewed payment systems primarily as neutral conduits for existing transactions, recent evidence indicates that payment innovations have measurable real economic effects. Patnam and Yao (2020) exploit the Indian demonetization shock to show that mobile money adoption increases household resilience, enabling consumption smoothing during periods of cash scarcity. Chodorow-Reich et al. (2019) provide the macroeconomic complement, estimating that the demonetization reduced GDP by approximately two percentage points - a reminder that disruptions to the payment system carry non-trivial aggregate costs. Crouzet et al. (2023) document a related technology adoption mechanism using the same shock: Indian merchants responded to the temporary cash contraction by persistently adopting electronic wallets, and the long-run adoption response was amplified by coordination complementarities, with adoption estimated to be 45% lower in their absence. These complementarities are directly analogous to the network externalities central to our analysis of BLIK.

More directly relevant to the market creation hypothesis, Agarwal et al. (2019) find that the introduction of QR-code payments in Singapore was associated with an 8.9% increase in business creation, concentrated among small, consumer-facing firms, though Van Hove (2022) subsequently argues that the observed increase in card sales cannot have been driven by the QR code technology and is more plausibly explained by the concurrent rise in contactless card adoption, raising questions about the causal identification in that study. Agarwal et al. (2024) provide complementary causal evidence from India’s 2016 demonetization: digital payment adoption is associated with an increase in aggregate consumer spending, attributable in part to a reduced psychological cost of transacting digitally relative to cash. Agarwal et al. (2025) document analogous patterns in the context of fintech adoption in China. Beyond Asia, Krivosheya (2020) documents that various types of financial innovations, i.e. mobile banking applications, internet banking, and e-wallets, are positively associated with the frequency of cashless payments among Russian consumers. However, Van Hove (2021) demonstrates that these associations reflect correlation rather than causation: Krivosheya’s estimation strategy does not isolate the impact of innovation adoption from the concurrent socioeconomic and behavioral characteristics of early adopters. This methodological concern, the difficulty of constructing credible counterfactuals for individual-level adoption studies, directly motivates the quasi-experimental approach employed in the present study, which exploits the market-level introduction of a payment standard to construct such a counterfactual at the aggregate level.

2.2. Technology Diffusion, Adoption Dynamics, and Payment Innovation

The adoption trajectory of new payment technologies can be understood through the lens of classical diffusion theory. Rogers (2003) identifies five adopter categories along an S-shaped cumulative adoption curve, where the transition from "early adopters" to the "early majority" represents a critical tipping point. For two-sided payment platforms, this tipping point is particularly demanding: consumer adoption and merchant acceptance must cross their respective thresholds simultaneously to generate self-reinforcing network effects (Van Hove et al., 2024). A systematic review of 106 empirical studies on digital financial transaction adoption identifies trust, perceived usefulness, and transaction cost reduction as the primary individual-level drivers of adoption across diverse institutional contexts (Kajol et al., 2022) which is a finding that maps directly onto the supply- and demand-side mechanisms examined in our macroeconomic analysis.

The Bass diffusion model (Bass, 1969) formalizes this process by decomposing adoption into innovation-driven (external influence) and imitation-driven (internal, network influence) components. In the context of payment systems, the innovation coefficient captures

the initial attractiveness of the technology’s functional advantages, such as the speed, security, reduced friction, while the imitation coefficient reflects the network externalities that accelerate diffusion once a critical mass of users and merchants is established. Recent empirical work on BLIK confirms that the system’s network value scales with the product of active users and accepting merchants rather than with user numbers alone, yielding a faster-than-linear growth pattern that is consistent with the S-curve dynamics predicted by diffusion theory (Van Hove et al., 2026).

A complementary framework for interpreting diffusion outcomes is the Multi-Level Perspective (MLP) of Geels (2002), which situates technology adoption within a dynamic interplay between innovation niches, incumbent socio-technical regimes, and the macro-level landscape. Lepoutre and Oguntoye (2018) apply the MLP to explain the differential emergence of mobile money systems in Kenya and Nigeria, two economies with comparable levels of economic development and mobile phone penetration, and find that Kenya’s rapid M-Pesa adoption and Nigeria’s near-failure reflect idiosyncratic differences in institutional conditions and ecosystem readiness rather than technology attributes per se. The key mechanism is that network externalities can only generate self-reinforcing adoption dynamics when the broader regime-level conditions, such as regulatory clarity, banking sector coordination, and consumer trust in digital channels, are sufficiently aligned. Applied to BLIK, the lesson from Lepoutre and Oguntoye (2018) suggests that Poland’s concentrated banking sector, early interchange fee regulation, and established mobile banking infrastructure constituted the regime-level preconditions that allowed BLIK’s network externalities to translate into the rapid diffusion observed empirically.

This diffusion perspective is central to interpreting our empirical results. BLIK’s growth from zero to 18.5 million active users (approximately 58 per 100 adults) within a decade, combined with its expansion from ATM withdrawals to e-commerce dominance, exhibits the hallmarks of a technology that has passed its tipping point. The widening gap between Poland and its synthetic counterfactual in the post-treatment period is accordingly expected to be non-linear, as network externalities compound over time - a pattern we examine empirically in Section 6.

2.3. Overlays, Sovereign Rails, and the Digital Euro

The digital payments landscape is increasingly characterized by competition among three distinct architectural models, each with different implications for transaction costs, financial intermediation, and payment sovereignty.

The first is the *overlay model*, exemplified by Big Tech wallets (e.g., Apple Pay, Google Pay). These systems tokenize credentials from existing card accounts to reduce friction at the checkout while maintaining the underlying interchange fee structure. Recent literature argues that this model can function as a rent-extraction mechanism, leveraging the Data

Network Activities loop to entrench incumbent platforms without reducing the cost of financial intermediation (Parlour & Rajan, 2025; Parlour, Rajan, & Zhu, 2022).

The second is the *sovereign rail model*, encompassing both centrally mandated instant payment infrastructures (such as Brazil’s Pix and India’s UPI) and bank-consortium domestic schemes (such as Poland’s BLIK, Spain’s Bizum, and Sweden’s Swish). By enabling direct account-to-account (A2A) settlement, these systems bypass the global card networks, decoupling domestic transaction costs from international interchange fee structures. This architectural distinction has implications for monetary policy transmission and competition in retail banking: sovereign A2A systems can increase deposit mobility by making account switching more seamless, potentially intensifying competition among deposit-taking institutions (Bech & Garratt, 2023; Parlour, Rajan, & Walden, 2022). Within this category, an important governance distinction separates centrally operated public utilities (Pix, UPI) from commercially governed bank-consortium standards (BLIK, Bizum). As discussed in Section 1, the former operate under mandatory participation and zero or near-zero merchant fees, while the latter retain commercial fee structures - an intermediate governance form that may prove more relevant as a model for European payment architecture.

The third emerging model is the *central bank digital currency* (CBDC). The European Central Bank’s digital euro project, currently in its preparation phase with a potential first issuance targeted for 2029 (European Central Bank, 2023a), represents the most advanced CBDC initiative in a major advanced economy. The digital euro is conceived as a retail CBDC that would complement cash and existing electronic payment instruments, offering a pan-European payment option that currently does not exist as no single electronic payment method covers the entire euro area (European Central Bank, 2025). In economies that already have an established domestic mobile payment scheme, the introduction of a CBDC may produce both additional substitution effect (displacing a share of incumbent mobile payment volume) and further market creation, as the CBDC reaches transaction segments where existing schemes have not yet achieved critical mass. Whether the net effect of CBDC adoption resembles the market-creation dynamics documented in this paper or reduces primarily to inter-scheme substitution is an empirical question whose resolution will depend critically on the institutional context of the adopting economy. The evidence presented in this paper provides a benchmark for calibrating such expectations, as it quantifies the market-creation potential of a sovereign domestic payment standard in a European context.

2.4. Evidence from Pix and UPI

The market creation hypothesis tested in this paper finds parallel support in the recent trajectories of Brazil and India. Both cases illustrate the dual nature of sovereign payment

systems: they simultaneously cannibalize existing digital flows and create genuinely new transaction volume, making the decomposition of these two channels, the central empirical task of the present paper a critical question.

In Brazil, Duarte et al. (2022) document that Pix both substituted existing digital instruments by displacing card transactions and traditional bank transfers and expanded the market by bringing approximately 50 million previously cash-only consumers into the digital payment ecosystem. The mandatory participation of large banks was instrumental in kickstarting the network effects that drove this rapid diffusion. At the macroeconomic level, non-cash retail transactions per capita in Brazil rose from 153 in 2013 to 790 by 2024, while ATM cash withdrawals declined by approximately 73% over the same period (Lobo et al., 2025). Beyond the aggregate payment shift, recent work has documented systemic effects: Sarkisyan (2025) shows that Pix adoption increased deposit market competition, while Gomes and Silva (2024) find that it diversified local credit market structures.

In India, the Unified Payments Interface has similarly reshaped the domestic monetary landscape. Launched in 2016, UPI processed 16.7 billion transactions in December 2024 alone (National Payments Corporation of India, 2026), accounting for approximately 80% of all retail digital payments (Reserve Bank of India, 2024). An important institutional distinction separates UPI from the BLIK model: UPI operates as a public utility offering a zero discount rate for merchants, which maximizes adoption speed but raises long-term sustainability questions (Aurazo & Gasmi, 2024). The Polish bank-consortium model retains commercial fee structures, representing an intermediate governance form between fully state-subsidized systems and rent-extracting private card overlays. Complementing the aggregate evidence, Dubey and Purnanandam (2025) provide causal estimates showing that UPI adoption increased household income and small-business activity, with effects concentrated among financially weaker households, -suggesting that the market-creation channel operates partly through a financial inclusion mechanism. The interoperability design of UPI offers a further lesson: Copestake et al. (2025) find that when two previously fragmented payment networks became interoperable on UPI, overall digital payment usage rose by more than 50%, driven by regions where fragmentation had been most severe.

A crucial difference between these cases and the present study is methodological. The evidence for Pix and UPI relies primarily on descriptive trajectories, event studies, or micro-level adoption analyses. The present paper contributes a macro-level counterfactual scenario constructed via the synthetic control method that allows for a direct estimate of the aggregate market-creation effect attributable to the payment innovation, net of concurrent policy interventions.

2.5. Two-Sided Markets and Network Effects

The theoretical underpinning of our analysis draws on the two-sided markets literature. While standard network growth rules (e.g., Metcalfe’s law) were developed for one-sided communication networks, Van Hove et al. (2024) argue that for platforms subject to indirect network effects, value creation scales with the product of the two network sides rather than with the square of total users. Van Hove et al. (2026) operationalize and test this scaling law using high-frequency BLIK data, finding strong empirical support: the system’s value proxy grows with the product of active users and accepting merchants ($n_1 \times n_2$). This non-linear growth trajectory, documented separately for point-of-sale and online networks, explains why sovereign payment systems can achieve accelerating adoption and rapid cash displacement once a critical mass tipping point is crossed. Błach and Klimontowicz (2021) provide complementary firm-level analysis of the institutional and organizational determinants of BLIK’s success within this two-sided market framework.

The importance of network externalities in payment markets is further underscored by field evidence from other contexts. Higgins (2024) exploits the Mexican government’s exogenous rollout of debit cards to poor households and finds important supply-side spillovers: small retailers adopted POS terminals in response, which in turn raised card adoption among other consumers by 21 percentage points. It is a sequence that mirrors the demand-supply feedback loop central to our analysis. Similarly, Carbó-Valverde et al. (2012) document a significant feedback loop in the Spanish card market: a 1% increase in a bank’s POS terminals raises its cardholder market share by 1.8%, while a 1% increase in cardholders produces a 2.7% increase in merchant demand for terminals. These findings confirm that indirect network externalities are not merely theoretical constructs but empirically consequential forces that shape the adoption and competitive dynamics of payment platforms. In the BLIK context, the widening gap between Poland and its synthetic counterfactual can be interpreted as the macroeconomic manifestation of precisely these compounding cross-side effects.

3. Institutional Background

3.1. The Polish Payment Landscape before BLIK

Prior to BLIK’s launch, the Polish payment landscape exhibited characteristics typical of post-transition economies: consumer adoption of card payments lagged behind Western Europe despite a well-developed banking infrastructure. In 2014, cash accounted for approximately 60% of point-of-sale transactions (Polasik et al., 2025). As documented by those authors, this digitization gap was sustained by a two-sided market failure. On one side it was high Merchant Service Charges (MSC) that suppressed merchant acceptance of

cards, on the other consumers had limited incentive to adopt cards given the constrained acceptance network. A key factor was the level of interchange fees: until 2014, Poland had among the highest interchange fee rates in Europe, which directly inflated merchant costs and discouraged terminal deployment. Poland’s banking sector, however, possessed a structural characteristic that would prove consequential for payment innovation: a relatively high degree of concentration, with the six largest commercial banks collectively serving the vast majority of retail customers. This concentration, combined with an established mobile banking infrastructure and high smartphone penetration, created the institutional preconditions for coordinated action that the MLP framework identifies as necessary for a payment innovation to achieve critical mass (Lepoutre & Oguntoye, 2018).

The years immediately preceding BLIK’s launch saw two important regulatory interventions. First, in January 2014, Poland introduced a statutory cap on interchange fees at 0.5% of transaction value, subsequently reduced to 0.2% for debit and 0.3% for credit cards in January 2015, ahead of the EU-wide Interchange Fee Regulation that took effect in December 2015 (Polasik et al., 2025). This reduction lowered the cost of card acceptance for merchants and began to expand the POS terminal network. Second, and crucially for the identification strategy employed in this paper, the interchange fee regulation preceded and is therefore independent of the BLIK treatment. In the synthetic control framework, the effects of the interchange fee reform are absorbed into the pre-treatment fit (2000-2014), as both Poland and its donor pool were subject to comparable regulatory trends during this period.

3.2. The Introduction and Evolution of BLIK

The intervention analyzed in this study is the launch of BLIK in February 2015. BLIK was established as a sovereign domestic payment scheme and system owned by Polski Standard Płatności (PSP), that was a consortium of the six largest commercial banks in Poland. Unlike “overlay” wallets (e.g., Apple Pay, Google Pay) that tokenize existing card rails, BLIK operates as an independent domestic payment standard that routes transactions through Poland’s existing banking and clearing infrastructure rather than through global card networks.

Functionally, the system operates on a code-based model: users generate a temporary six-digit code within their mobile banking application, which is then entered at a point-of-sale terminal, ATM, or e-commerce checkout. This architecture creates indirect network effects: BLIK’s value scales with the product of active users and accepting merchants ($n_1 \times n_2$) rather than with the square of users alone (Van Hove et al., 2026). By end-2024, 21 banks participated in the system, representing virtually all retail banks in the country. BLIK had reached 18.5 million active users (approximately 58 per 100 adults) and accounted for approximately 67% of the value of all domestic online transactions and

6.3% of all cashless transactions at the point of sale (Van Hove et al., 2026).

BLIK's expansion has proceeded in distinct phases, each extending the system into new transaction domains. At launch in 2015, BLIK supported ATM withdrawals and e-commerce payments. Peer-to-peer (P2P) transfers, settled in real time through Express Elixir, Poland's instant payment system operated by KIR (Górka, 2025), were added shortly thereafter. The most consequential expansion for point-of-sale adoption was the introduction of contactless payments in July 2021, which operate in cooperation with Mastercard. Prior to this, BLIK users had to enter a six-digit code at the terminal (a feature that remains available) that was an approach slower than contactless card payments. The contactless feature eliminated this friction, and POS transaction volumes accelerated markedly: by Q4 2024, BLIK processed 154.6 million POS transactions in a single quarter (Van Hove et al., 2026). This phased expansion is relevant for interpreting the time-varying treatment effect documented in Section 6: the widening of the gap between Poland and its synthetic counterfactual from 2021 onward is consistent with the contactless feature compressing the adoption S-curve at the point of sale (cf. Viardot et al., 2023).

3.3. The Cashless Payment Support Program (CPSP)

A concurrent policy intervention that must be separated from the BLIK technology effect is the Cashless Payment Support Program (CPSP). Polasik et al. (2025) characterize Polish payment policy of this period as a "carrot and stick" approach: the interchange fee regulation (the "stick") preceded BLIK's launch and is absorbed into the pre-treatment period, but the CPSP was introduced during the treatment window itself.

The CPSP, formally launched in January 2018 and operated through the Cashless Poland Foundation, covered terminal installation and leasing costs for up to 12 months for participating micro-merchants. The program was designed to address the supply-side bottleneck in the two-sided market: even with reduced interchange fees, many small merchants lacked the resources or incentive to invest in terminal infrastructure. By end-2024, approximately 48% of all operating POS terminals in Poland had been introduced under CPSP subsidies (Polasik et al., 2025). Because POS terminal density is a structural precondition for cashless payments—including but not limited to BLIK—attributing the CPSP-driven infrastructure expansion to the BLIK technology would inflate the estimated treatment effect. The two-stage decomposition described in Section 5 is designed to address precisely this confound.

3.4. European Domestic Mobile Payment Schemes

Tables 1 and 2 situate BLIK within the broader landscape of European domestic mobile payment schemes launched between 2000 and 2024, classified by governance architecture. Of the 19 active schemes identified, 13 are operated by bank consortia, confirming

that the consortium model has proved the dominant organizational form for surviving domestic payment standards in Europe. Notably, all telco-led schemes and corporate-led ventures have collapsed, suggesting that the commercial sustainability of domestic payment infrastructure depends critically on bank participation and the associated trust infrastructure. Central-bank-operated schemes represent an emerging fourth category, with qvik (Hungary) and RoPay (Romania) both launched in 2024, though their long-run viability remains unproven. The ongoing consolidation wave, including Giropay's closure in Germany, Paylib's replacement by Wero in France, Vipps and MobilePay's merger into Vipps MobilePay, and the pending rebranding of Payconiq by Bancontact as Bancontact Pay, underscores both the competitive pressure that pan-European initiatives are placing on national schemes and a broader trend toward larger, multi-country platforms rather than narrow domestic ones.

Table 1: Active European Domestic Mobile Payment Schemes

Scheme	Country	Launch	Operator
<i>Bank consortium</i>			
Swish	Sweden	2012	Getswish AB (6 Swedish banks)
BLIK	Poland	2015	Polski Standard Płatności (PSP)
MB WAY	Portugal	2015	SIBS (interbank consortium)
Bizum	Spain	2016	30+ Spanish banks
TWINT	Switzerland	2017	TWINT AG (6 Swiss banks)
IRIS ^a	Greece	2017	DIAS Interbanking Systems S.A.
BANCOMAT Pay ^b	Italy	2019	BANCOMAT S.p.A. (interbank)
Blink ^c	Bulgaria	2021	BORICA AD (national payment operator)
Vipps MobilePay	Norway, Denmark, Finland	2022	Vipps MobilePay AS
BLIK SK ^d	Slovakia	2023	Polski Standard Płatności (PSP)
BLIK RO ^f	Romania	2024	Polski Standard Płatności (PSP)
Cvak	Czechia	2024	Czech bank consortium (Air Bank-led)
Wero	Belgium, France, Germany	2024	EPI (16-bank European consortium)
<i>Single bank</i>			
Tikkie	Netherlands	2016	ABN AMRO
KUIK ^e	Albania	2019	MPAY sh.p.k. / Raiffeisen Bank Albania
<i>Central bank / public authority</i>			
qvik	Hungary	2024	National Bank of Hungary
RoPay	Romania	2024	TransFonD / National Bank of Romania
<i>Independent fintech</i>			
Satispay	Italy	2015	Satispay S.p.A.
Bluecode	Austria, Germany	2015	Blue Code International AG

^a IRIS is operated by DIAS Interbanking Systems S.A., Greece’s national automated clearing house founded in 1989 by 14 commercial banks; the Bank of Greece is also a shareholder. IRIS processed 57.3 million transactions in 2024 (+137% year-on-year) and mandatory merchant acceptance was introduced in December 2025. DIAS is a member of EMPSA and joined the EuroPA cross-border instant payment network in 2025.

^b BANCOMAT Pay is the successor to Jiffy (2014–2018), originally operated by SIA before integration into the BANCOMAT interbank infrastructure.

^c Blink is Bulgaria’s national instant payment brand, operated by BORICA AD—a payment infrastructure company owned by 28 Bulgarian banks and the Bulgarian National Bank. The first live transaction was processed in November 2021; Blink P2P (mobile-number lookup) was added in 2023. By 2024, Blink processed 22.6 million payments annually, reaching over 90% of Bulgarian bank accounts.

^d BLIK SK is the successor to VIAMO, a scheme launched in Slovakia in 2013 by a Slovak bank consortium. Following PSP’s acquisition of VIAMO in 2022, the scheme was relaunched as BLIK SK in 2023, marking BLIK’s first international expansion.

^e KUIK is operated by MPAY sh.p.k. in partnership with Raiffeisen Bank Albania. Introduced in May 2019 as the successor to mpay (2013–2021), it requires a Raiffeisen Bank account and functions as a peer-to-peer instant transfer service.

^f BLIK Romania S.A. received authorization from the National Bank of Romania on October 1, 2024, marking PSP’s second international expansion after Slovakia. At authorization, the operator indicated an initial focus on e-commerce channels; consumer-facing deployment is pending bank partnerships.

Table 2: Defunct European Domestic Mobile Payment Schemes

Scheme	Country	Launch	Closed	Fate
<i>Bank consortium</i>				
iDEAL ^a	Netherlands	2005	2023	Acquired by EPI; transitioning to Wero
Giropay	Germany	2006	2024	Shut down
Digicash	Luxembourg	2012	2017	Acquired by Payconiq
Swipp	Denmark	2013	2017	Migrated to MobilePay
Paylib	France	2013	2024	Replaced by Wero
MobilePay	Denmark, Finland	2013	2022	Merged into Vipps MobilePay
VIAMO	Slovakia	2013	2022	Rebranded as BLIK SK (PSP)
Payconiq	Belgium, Lux.	2014	2026	Rebranded as Bancontact Pay ^b
Jiffy	Italy	2014	2018	Integrated into BANCOMAT Pay
Paym	United Kingdom	2014	2023	Shut down
paydirekt	Germany	2015	2021	Merged into Giropay
Vipps	Norway	2015	2022	Merged into Vipps MobilePay
Kwitt	Germany	2016	2021	Merged into Giropay
Payconiq NL	Netherlands	2018	2022	Redirected to iDEAL
<i>Single bank</i>				
Pivo	Finland	2012	2023	Shut down (OP Financial Group)
<i>Independent fintech</i>				
mCASH	Norway	2012	2017	Absorbed into Vipps
mpay	Albania, Kosovo	2013	2021	Mission completed; succeeded by KUIK
<i>Telco consortium</i>				
SimPay	Pan-European	2003	2005	Collapsed pre-launch
mpass	Germany	2011	2016	Shut down
<i>Corporate / bank-retail hybrid</i>				
Mobipay	Spain	2002	2009	Shut down (Telefónica + banks)
Yapital	Germany, Austria	2013	2016	Shut down (Otto Group)
Lyf Pay	France	2017	2026	Shut down (BNP Paribas + Carrefour)

^a iDEAL was initially launched as an e-commerce bank-transfer protocol and later evolved into a mobile-first account-to-account payment app. It was acquired by the European Payments Initiative in 2023 and is transitioning to the Wero infrastructure; the iDEAL brand continues to operate in the Netherlands during this transition period.

^b The Payconiq by Bancontact app is not being shut down but rebranded as Bancontact Pay in spring 2026. The corporate entity simultaneously rebrands as Bancontact Company and acts as a distributor of Wero in Belgium, making Bancontact Pay and Wero parallel rather than competing offerings.

Note: “Defunct” denotes schemes that have ceased independent operation, regardless of whether a successor service exists. Schemes rebranded or absorbed into active platforms (e.g., MobilePay and Vipps → Vipps MobilePay; Payconiq → Bancontact Pay) are classified as defunct at the point of brand discontinuation.

4. Data and Descriptive Statistics

4.1. Data Sources and Variable Construction

We construct a balanced panel of 22 European economies (Poland and 21 donor countries) spanning the period 2000-2024. The data are harmonized from three primary sources to ensure cross-border comparability.

The dependent variable (Y) is the value of electronic retail payments normalized by Household Final Consumption Expenditure (HFCE). For EU member states, data are drawn from the European Central Bank (ECB) Statistical Data Warehouse. For non-euro-area countries (Poland, Albania, Bosnia and Herzegovina), we use national central bank statistics. The Polish series includes both card transactions and BLIK mobile payment flows at POS terminals and in e-commerce, whereas donor countries are measured through card payment data only. This asymmetry is by design: since donor countries did not develop comparable mobile payment schemes, card payments constitute the entirety of their electronic payment activity. Including BLIK in the Polish outcome variable therefore measures the *incremental* digitization attributable to the mobile standard, i.e. the precise object of causal interest. As a consequence, our estimates should be interpreted as an upper bound of the pure technology effect, as they also capture any BLIK flows that may have displaced card transactions rather than cash (see Section 8 for further discussion).

The predictor set (X) comprises five covariates selected through a specification search over more than 1,000 candidate combinations to minimize the root mean squared prediction error (RMSPE) for the pre-treatment period (2000-2014), following Abadie (2021). GDP per capita (PPP, constant 2017 international \$; World Bank) controls for the baseline income level that drives payment card adoption. The IMF Financial Development Index (Svirydzenka, 2016) captures the depth and efficiency of financial institutions and markets, which is a direct determinant of the infrastructure available for electronic transactions. Tertiary education enrollment (% gross; World Bank) proxies for the human capital and digital literacy that condition technology adoption rates (Rogers, 2003). Internet penetration (% of population; ITU) measures the digital infrastructure prerequisite for both e-commerce and mobile payment usage. Finally, the Rule of Law Index (World Bank WGI) controls for the institutional quality and contract enforcement environment that shapes consumer trust in digital financial services (Kajol et al., 2022). Table 3 summarizes the predictor set.

Table 3: Predictor Variables for the Synthetic Control Estimation

Variable	Source	Rationale
GDP per capita (PPP, const. 2017 int. \$)	World Bank	Baseline income driving card adoption
Financial Development Index	IMF	Depth of infrastructure for electronic transactions
Tertiary education enrollment (% gross)	World Bank	Digital literacy conditioning adoption rates
Internet penetration (% of pop.)	ITU	Prerequisite for e-commerce and mobile payments
Rule of Law Index	World Bank WGI	Contract enforcement and trust in digital services

4.2. Donor Pool Construction and Exclusion Criteria

The validity of the synthetic counterfactual relies on a donor pool that shares structural similarities with the treated unit but remains unaffected by the intervention or similar idiosyncratic shocks. We begin with the universe of EU-27 and EFTA economies and apply two exclusion criteria.

First, we remove countries that introduced domestic mobile payment standards prior to or concurrently with Poland, as their inclusion would violate the no-interference assumption. Denmark is excluded due to MobilePay (launched 2013) and Sweden due to Swish (launched 2012). Greece presents a partial exception: IRIS Payments was launched in 2020, five years after the Polish treatment date. We retain Greece in the donor pool because its scheme was introduced well after the pre-treatment window, its retail commerce penetration remains negligible relative to BLIK, and the optimization assigns it a low weight (5.8%).¹

Second, we exclude economies with insufficient reporting on HFCE or payment volumes throughout the full panel (e.g., Liechtenstein). Lithuania, which adopted the euro in January 2015, coinciding with the treatment date, is retained; its payment series exhibits no anomalous discontinuity attributable to the currency changeover, and the optimization assigns it zero weight, validating its inclusion post hoc.

The final estimation sample comprises 21 donor economies: Albania, Austria, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czechia, Estonia, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Romania, Slovakia, and Slovenia. The pool spans a range of income levels, financial development stages, and institutional contexts, but is anchored by CEE peers (Hungary, Czechia, Slovakia) that share Poland’s post-transition banking structure while lacking a sovereign payment standard during the sample period.

¹In 2024, IRIS Commerce recorded approximately 7.3 million e-commerce transactions (Van Hove et al., 2026, DIAS Annual Report 2024), compared to BLIK’s approximately 67% share of domestic online transaction value. The modest synthetic weight further limits any potential contamination of the counterfactual trajectory.

4.3. Synthetic Control Weights and Pre-Intervention Balance

The optimization produces a sparse weight vector: of the 21 donor economies, only five receive positive weight, while the remaining 16, including the Western European economies, which were retained to expand the permutation inference sample, receive zero weight. Table 4 reports the full weight vector.

Table 4: Country Weights in the Synthetic Poland Control Unit

Country	Weight	Country	Weight
Albania	0.071	Hungary	0.388
Austria	0.000	Ireland	0.000
Bosnia and Herzegovina	0.000	Italy	0.000
Bulgaria	0.124	Latvia	0.359
Croatia	0.000	Lithuania	0.000
Cyprus	0.000	Luxembourg	0.000
Czechia	0.000	Malta	0.000
Estonia	0.000	Romania	0.000
France	0.000	Slovakia	0.000
Germany	0.000	Slovenia	0.000
Greece	0.058		

Note: Weights are derived from the minimization of the RMSPE for the pre-treatment period (2000–2014). Countries are listed alphabetically.

The five positively weighted donors yield the following synthetic counterfactual:

$$\hat{Y}_{\text{POL}} = 0.388Y_{\text{HUN}} + 0.359Y_{\text{LVA}} + 0.124Y_{\text{BGR}} + 0.071Y_{\text{ALB}} + 0.058Y_{\text{GRC}} \quad (1)$$

The composition is economically interpretable. Hungary (38.8%) and Latvia (35.9%) jointly account for nearly three-quarters of the counterfactual. Both are post-transition EU member states with comparable GDP per capita trajectories, similar levels of financial development, and payment markets that remained card-based throughout the sample period, with no domestic mobile payment standard of comparable scale. Bulgaria (12.4%) and Albania (7.1%) contribute as lower-income anchors that help match the earlier, slower phase of Poland’s digitization trajectory during 2000–2008, when electronic payment adoption was still incipient. Greece (5.8%) enters at a modest weight, reflecting its intermediate position between CEE and Western European payment structures. Western European donors, such as Austria, France, Germany, Ireland, and Italy, received zero weights, consistent with the expectation that economies with structurally higher card penetration, deeper financial markets, and longer histories of electronic payment adoption

represent poor counterfactuals for a post-transition economy.

Table 5 reports the pre-intervention balance between Poland and its synthetic counterpart.

Table 5: Pre-Intervention Predictor Means (2000–2014)

Predictor	Poland	Synthetic Poland
GDP per capita (PPP, '000 Int. \$)	24.41	24.40
Financial Development Index (IMF)	0.404	0.366
Internet Access (% of pop.)	43.7%	43.7%
Tertiary Education (% gross)	66.6%	61.5%
Rule of Law Index (WGI)	0.625	0.625

The optimization achieves a close fit across all five predictor dimensions. GDP per capita is matched to within \$10 (24.41 vs. 24.40 thousand), and internet penetration is identical at 43.7%, indicating that the synthetic unit replicates Poland’s income level and digital infrastructure almost exactly. The Rule of Law Index is also matched precisely (0.625), suggesting comparable institutional environments for digital financial services. The two predictors with the largest residual discrepancies, the Financial Development Index (0.404 vs. 0.366, a gap of 0.038) and tertiary education enrollment (66.6% vs. 61.5%, a gap of 5.1 percentage points), both show the synthetic unit slightly below Poland, which would, if anything, bias the pre-treatment fit toward underpredicting Poland’s payment trajectory and thus produce a conservative estimate of the post-treatment gap. The close overall balance, combined with the low pre-treatment RMSPE of approximately 0.24 percentage points of HFCE reported in Section 6, supports the conclusion that the synthetic counterfactual provides a credible approximation of Poland’s payment trajectory in the absence of the BLIK innovation.

5. Econometric Framework

To isolate the “market creation” effect of the mobile payment system from concurrent infrastructure subsidies, we employ a nested two-stage identification strategy.

5.1. Stage 1: Penalized Synthetic Control Method

We first estimate the aggregate treatment effect using the synthetic control method (SCM). We construct a counterfactual outcome Y_{it}^N as a weighted linear combination of the donor pool J . The optimal weight vector $W^* = (w_2, \dots, w_{J+1})'$ is chosen to minimize the distance between the predictor vectors X_1 and X_0W :

$$\min_W \|X_1 - X_0W\|_V = \sqrt{(X_1 - X_0W)'V(X_1 - X_0W)} \quad (2)$$

where V is a symmetric, positive-definite matrix assigning relative importance to each predictor. We tested over 1,000 specifications to select the predictor set defined in Section 4. The resulting weight vector and pre-intervention predictor balance are reported in Tables 4 and 5 in Section 4.

5.2. Stage 2: Orthogonalizing the Supply-Side Shock

A confounding factor is the Cashless Payment Support Program (CPSP), which subsidized POS terminals starting in 2018. To adhere to the exclusion restriction, we must purge the portion of the treatment effect attributable to this supply-side intervention.

5.2.1. Counterfactual POS Estimation (ARIMAX)

We model the natural growth of POS infrastructure using an ARIMAX(p, d, q) framework. We train the model on the pre-subsidy sample ($t \in [2004Q1, 2017Q4]$). The data generating process is defined as:

$$\Delta y_t = \mu + \sum_{i=1}^2 \phi_i \Delta y_{t-i} + \beta' \mathbf{Z}_t + \epsilon_t \quad (3)$$

where y_t is the stock of POS terminals and $\mathbf{Z}_t$ is a vector of exogenous controls: unemployment rate, GDP per capita, and internet penetration. The difference between the observed terminal count and the model forecast for 2023 represents the specific contribution of the subsidy ($\Delta \text{POS}_{\text{CPSP}} \approx 375,000$ units).

5.2.2. Elasticity of Infrastructure

To translate $\Delta \text{POS}_{\text{CPSP}}$ into transaction value, we estimate a Two-Way Fixed Effects (TWFE) panel model on 42 countries (2001–2021). We explicitly test for diminishing marginal utility via an interaction term:

$$Y_{it} = \alpha_i + \lambda_t + \beta_1 T_{it} + \beta_2 (T_{it} \times G_{it}) + \gamma' \mathbf{X}_{it} + \epsilon_{it} \quad (4)$$

where T_{it} is terminals per capita and G_{it} is GDP per capita. Consistent with saturation theory, we find a negative interaction term ($\beta_2 < 0$). Applying the marginal effect calculated for Poland's specific GDP level in 2023 (≈ 0.072), we estimate the supply-side impact to be $\hat{\tau}_{\text{CPSP}} = 2.7\%$ of HFCE.

5.3. Net Treatment Effect and Bounds Analysis

The net causal impact of the mobile payment innovation is defined as the residual of the aggregate SCM gap and the supply-side adjustment. Using the 2023 endpoint as the reference for the decomposition (where the CPSP elasticity model is fully identified):

$$\hat{\tau}_{BLIK} = \hat{\tau}_{SCM} - \hat{\tau}_{CPSP} = 5.8\% - 2.7\% \approx 3.1\% \quad (5)$$

Applying the bias-corrected SCM estimate reduces the aggregate gap to 5.5 percentage points at the 2023 endpoint, yielding a bias-corrected net technology effect of approximately 2.8 percentage points. Extending the analysis to 2024, the aggregate gap widens to approximately 9.0 percentage points (classic estimate) or 8.6 percentage points (bias-corrected), reflecting continued BLIK diffusion and the acceleration associated with the contactless payment feature introduced in 2021.

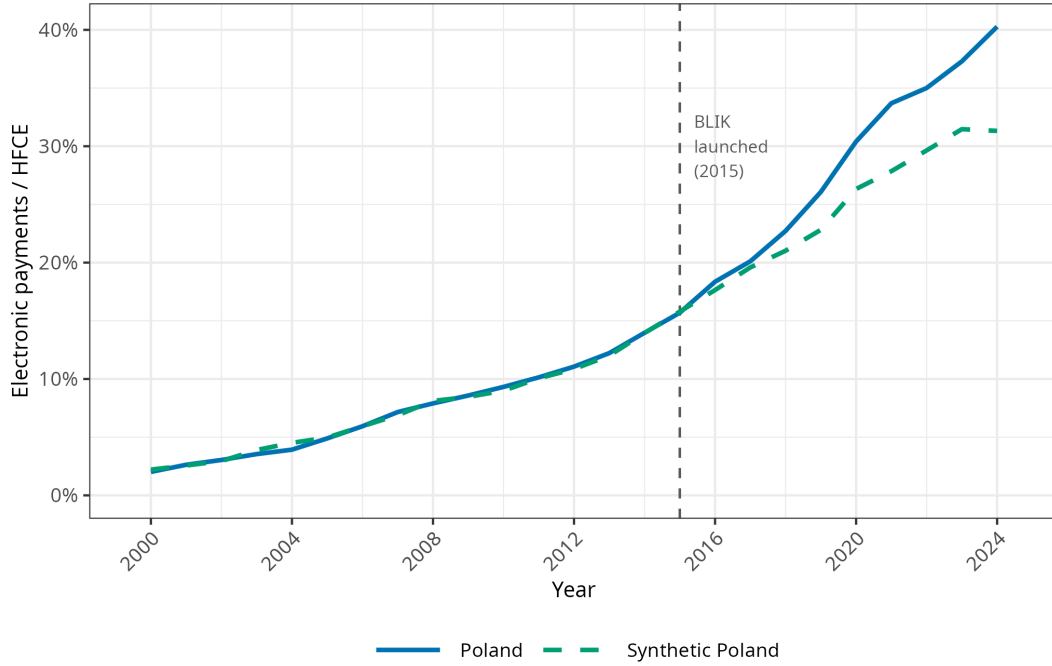
To assess robustness of the decomposition, we conduct a bounds analysis. Under the conservative assumption that the CPSP was 50% more effective than our elasticity model predicts ($\delta = 1.5$, implying a supply-side contribution of 4.1 pp), the net technology effect remains positive and economically meaningful at approximately 1.7 percentage points (classic). This finding is consistent with the hypothesis that the mobile payment standard generated digitization beyond what infrastructure subsidies alone could explain.

6. Empirical Results

6.1. Pre-Intervention Fit and Counterfactual Validity

The validity of the synthetic control estimator rests on its ability to reproduce the trajectory of the treated unit prior to the intervention. As shown in Figure 1, the synthetic counterfactual tracks the actual Polish series with high precision throughout the pre-treatment period (2000–2014). The pre-treatment RMSPE of approximately 0.24 percentage points of HFCE confirms a close pre-intervention fit: absent the specific payment shocks introduced in 2015, the trajectory of electronic payments in Poland would plausibly have followed a path similar to the weighted average of its CEE and Southern European peers.

Figure 1: Electronic Payments in Poland vs. Synthetic Poland, 2000–2024



Note: The dashed vertical line marks the BLIK treatment date (2015). Synthetic Poland (dashed) is the weighted average of Hungary (38.8%), Latvia (35.9%), Bulgaria (12.4%), Albania (7.1%), and Greece (5.8%). The outcome variable is the value of electronic retail payments as a percentage of Household Final Consumption Expenditure (HFCE).

6.2. The Aggregate Impact of Payment Innovation

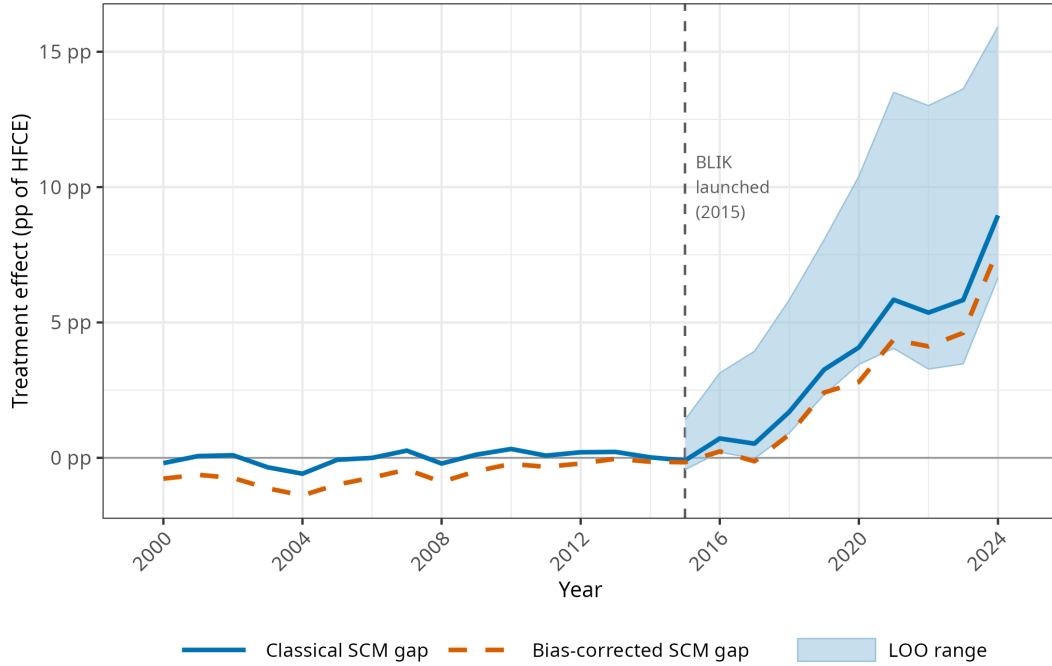
Following the introduction of BLIK in 2015, a distinct divergence emerges between the two series. While both Poland and its synthetic counterpart show a trend toward payment digitization, the Polish trajectory accelerates progressively relative to the counterfactual from 2016 onward, with a pronounced step-up from 2018–2019 as network externalities compound.

The aggregate gap widens monotonically over the post-treatment period. By 2023, the estimated treatment effect reaches approximately **5.8 percentage points** of Household Final Consumption Expenditure (classic estimate) or **5.5 percentage points** (bias-corrected), continuing to widen to approximately **8.9 pp** (classic) or **8.6 pp** (bias-corrected) by end-2024. These figures represent the gross excess volume of digital transactions in Poland relative to what regional trends and fundamental economic drivers would predict in the absence of the mobile payment innovation.

Figure 2 plots the treatment gap series together with the bias-corrected estimate and the leave-one-out (LOO) sensitivity corridor. The corridor spans the range of gap estimates obtained by iteratively excluding each of the five positively weighted donors; its width reflects donor-composition uncertainty rather than statistical confidence intervals.

The classic and bias-corrected series track each other closely throughout, with the bias correction consistently reducing the gap by 1-2 pp, as expected under the regularization framework of Abadie (2021).

Figure 2: SCM Treatment Gap with Leave-One-Out Sensitivity Corridor, 2000–2024



Note: The solid line is the classic SCM gap (Poland minus Synthetic Poland); the dashed line is the bias-corrected estimate. The shaded band spans the minimum and maximum gap obtained by iteratively removing each of the five positively weighted donors. The dashed vertical line marks the BLIK treatment date (2015).

The LOO corridor in Figure 2 is quantified in Table 6, which reports the estimated treatment gap for each leave-one-out iteration across the full post-treatment period. The baseline model assigns substantial weights to Hungary (38.8%) and Latvia (35.9%); Table 6 assesses whether the aggregate gap is robust to the removal of any single high-weight donor.

Table 6: Leave-One-Out Sensitivity Analysis: Treatment Gap by Year (pp of HFCE)

Year	–HUN	–LVA	–BGR	–ALB	–GRC	Mean	Main	Corrected
2015	1.4	1.0	–0.2	–0.4	–0.4	0.3	–0.1	–0.2
2016	3.1	1.5	0.4	0.2	0.4	1.1	0.7	0.6
2017	3.9	0.7	0.2	0.0	0.1	1.0	0.5	0.4
2018	5.8	1.6	1.5	1.1	0.9	2.2	1.7	1.5
2019	8.0	3.3	3.0	2.6	2.3	3.9	3.3	3.1
2020	10.4	4.5	3.6	3.4	3.5	5.1	4.1	3.8
2021	13.5	4.0	5.6	5.3	5.0	6.7	5.8	5.5
2022	13.0	3.3	5.3	4.8	4.1	6.1	5.4	5.3
2023	13.6	3.5	6.3	5.2	4.1	6.6	5.8	5.5
2024	15.9	6.7	9.3	8.3	7.6	9.6	9.0	8.6

Note: Each column reports the estimated treatment gap (percentage points of HFCE) when the indicated donor country is removed and the synthetic control is re-estimated. –HUN, –LVA, –BGR, –ALB, and –GRC denote removal of Hungary, Latvia, Bulgaria, Albania, and Greece, respectively. “Mean” is the cross-iteration average. “Main” and “Corrected” are the baseline classic and bias-corrected SCM estimates using the full donor pool.

The range of estimates across iterations is informative about the sensitivity of the result. Excluding Hungary, which has the largest weight and recorded a decline in its electronic payment share in 2024 (39.97% to 38.54% of household consumption), reversing three consecutive years of growth, produces the highest gap estimate (13.6 pp in 2023), while excluding Latvia yields the lowest (3.5 pp). The cross-iteration mean (6.6 pp) exceeds the baseline estimate (5.8 pp), driven almost entirely by the Hungary-exclusion scenario: when Hungary is dropped, the remaining donors (dominated by Latvia and Bulgaria at higher combined weights) cannot fully replicate Poland’s pre-2015 trajectory, which upwardly biases the synthetic gap. The four remaining LOO iterations (excluding LVA, BGR, ALB, GRC) cluster tightly between 3.5 and 6.3 pp in 2023, confirming that the result is robust to removal of any single lower-weight donor. The direction and statistical significance of the post-2015 divergence are consistent across all iterations, confirming that the estimated gap reflects a generalized deviation from the CEE regional trend rather than an artifact of one particular control unit.

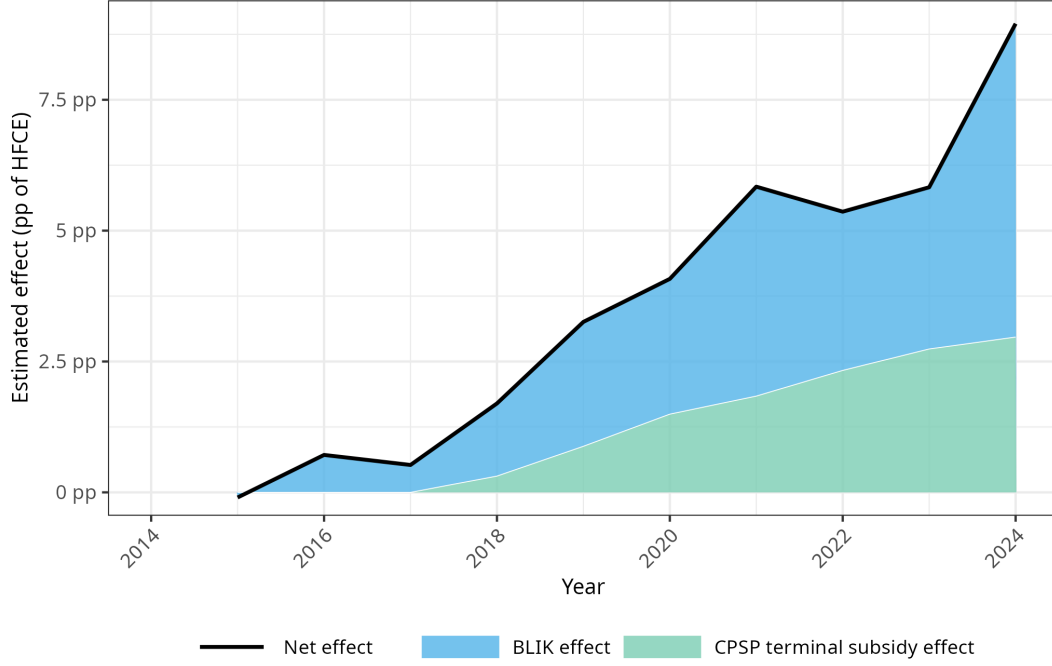
6.3. Effect Decomposition: Infrastructure Subsidy versus Technology Effect

A credible estimate of the mobile payment technology effect requires isolating it from a concurrent supply-side intervention. Polasik et al. (2025) characterize Polish payment

policy of this period as a "carrot and stick" approach: the interchange fee cuts (the stick) preceded BLIK's launch and are absorbed into the pre-treatment control, but the Cashless Payment Support Program (CPSP) was introduced as the carrot during the treatment window itself. The CPSP, formally launched in January 2018 and operated through the Cashless Poland Foundation, covered terminal installation and leasing costs for up to 12 months for participating micro-merchants. By end-2024, approximately 48% of all operating POS terminals in Poland had been introduced under CPSP subsidies. Because POS terminal density is a structural precondition for cashless payments, attributing the CPSP-driven infrastructure expansion to the BLIK technology would inflate the estimated treatment effect. The two-stage decomposition described in Section 5 is designed to address precisely this confound.

Figure 3 presents the year-by-year decomposition of the treatment gap into the CPSP terminal-subsidy component and the residual net technology effect. The two components are additive by construction (equation 3). The supply-side contribution grows steadily from 2018 onward as the CPSP program scales, reaching approximately 2.7 pp in 2023, while the net BLIK technology effect accounts for the remaining and larger share of the cumulative gap. The figure illustrates that the dominance of the technology channel over the infrastructure channel is a persistent feature of the post-treatment period rather than a late-sample artifact.

Figure 3: Decomposition of the Treatment Effect



Note: The stacked area shows the additive decomposition of the classic SCM gap. The CPSP component is derived from the Stage 2 TWFE elasticity applied to the ARIMAX-projected counterfactual terminal stock; the net BLIK component is the residual. The solid black line is the total estimated gap.

6.4. Further Robustness and Sensitivity Analysis

To further ensure that the estimated treatment effect is not an artifact of spurious correlations or the specific construction of the placebo distribution, we conduct two additional tests standard in the SCM literature.

6.4.1. In-Space Placebo Tests

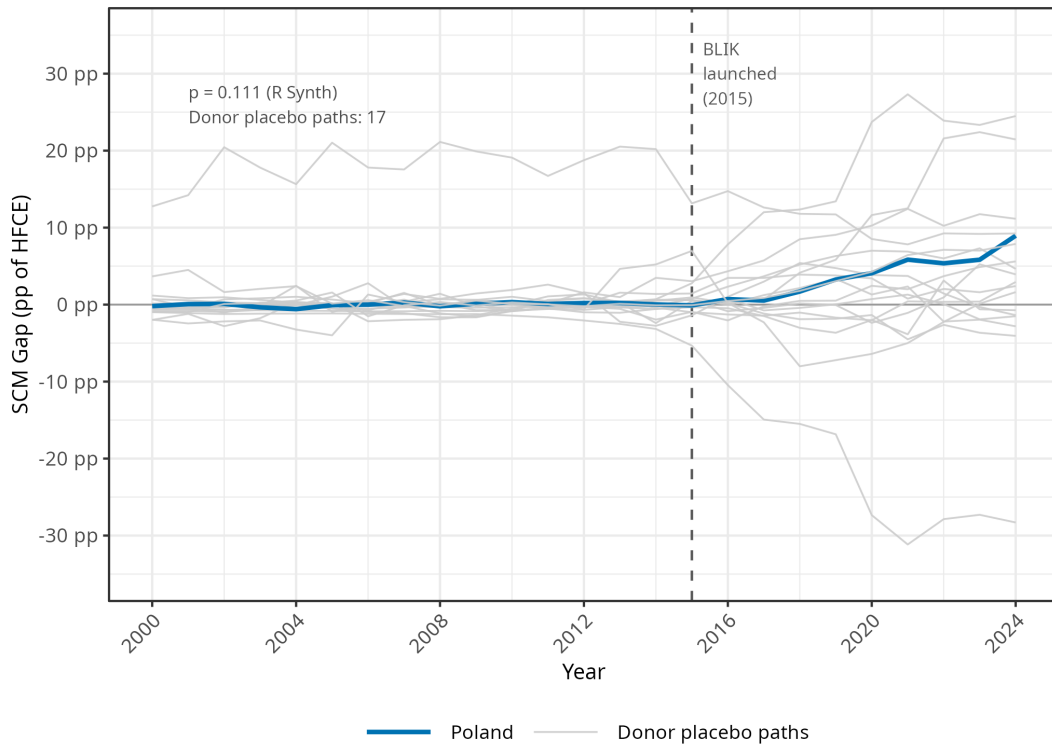
Following Abadie (2021), we conduct “in-space” placebo tests by reassigning the treatment intervention to every country in the donor pool. We generate synthetic control units for all 21 non-treated countries and compare the distribution of their post-2015 prediction errors (RMSPE) to that of Poland.

If the estimated gap were driven by random variation, Poland’s post-intervention RMSPE ratio would be expected to fall within the distribution generated by the donor countries. Under the classic (uncorrected) SCM, Poland’s permutation p -value is $p = 3/22 \approx 0.136$ (Poland plus 21 donors), as three donor units produce a higher post-to-pre RMSPE ratio in some post-treatment years. Under the bias-corrected SCM (`bcorrect(merge ridge)`), Poland achieves $p = 1/22 \approx 0.045$ from 2021 onward—the lowest among all 22 units—consistent with statistical significance at the 5% level under the Fisher randomization

inference framework. The inference therefore rests on the bias-corrected permutation test, which accounts for the regularization bias inherent in penalized SCM weights.

Figure 4 displays the in-space placebo paths. Poland’s gap (solid blue) is consistently positive and grows monotonically after 2015, while most donor placebo paths oscillate around zero or turn negative. Several donor paths do, however, exhibit larger positive deviations in some post-treatment years—a pattern reflected in the figure-level p -value of 0.111 under the R **Synth** specification, where approximately two of the 17 donor paths produce a more extreme post-to-pre RMSPE ratio than Poland in at least some years. Crucially, no comparable pre-2015 divergence is visible for Poland, supporting the causal interpretation that the structural break is linked to the BLIK launch rather than a pre-existing trend. The figure-level p -value should be read as indicative only; the permutation test over 22 units (yielding $p = 1/22 \approx 0.045$ under bias correction) is reported in the preceding paragraph.

Figure 4: SCM Gaps for Poland and Donor Countries, 2000–2024



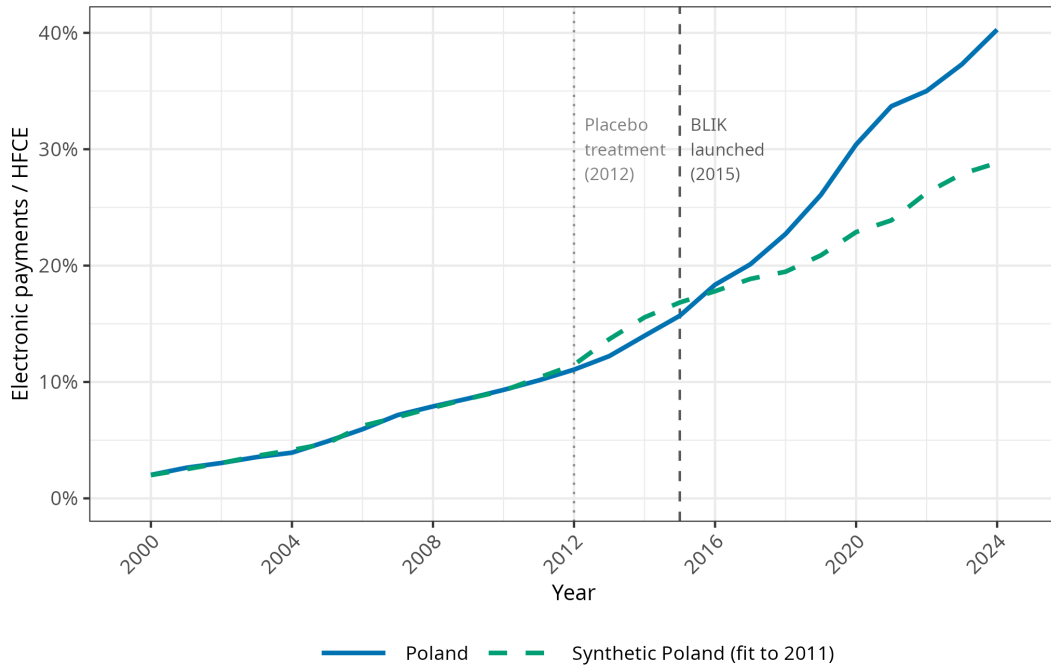
Note: Gray lines represent placebo gaps for donor countries; the figure reflects the R **Synth** replication over 17 donors for illustration. Poland’s gap (solid blue) is based on the Stata **allsynth** estimate. The Stata permutation test over 22 units (Poland + 21 donors) yields a classic $p = 3/22 \approx 0.136$ and a bias-corrected $p = 1/22 \approx 0.045$ from 2021 onward.

6.4.2. Temporal Placebo Tests

We further test for spurious correlations by shifting the intervention date artificially. We re-run the optimization assigning the treatment year to 2012 (pre-intervention). If our model is well-specified, Poland should not rise above the synthetic counterfactual between 2012 and 2015. The results confirm this: Poland does not exceed the placebo synthetic prior to the actual BLIK launch, supporting the causal interpretation that the structural break is linked to the introduction of the payment system rather than to any earlier trend.

Figure 5 displays the temporal placebo. Up to 2012, Poland and the synthetic counterfactual track closely. Between 2012 and 2015, however, the synthetic, now fitted only to 2000–2011, actually *overshoots* actual Poland, predicting a higher trajectory than was realized. Poland only surpasses the synthetic counterfactual after the actual BLIK launch in 2015. This pattern directly rules out a pre-existing upward trend as an explanation for the main result: had a structural break been present before 2015, Poland would have risen above the synthetic from 2012 onward. The opposite is observed.

Figure 5: Temporal Placebo Test: Poland vs. Synthetic Poland



Note: The dotted vertical line marks the false treatment date (2012); the dashed vertical line marks the actual BLIK launch date (2015). Between 2012 and 2015 the synthetic slightly overshoots actual Poland; Poland only surpasses the synthetic after 2015, confirming that the post-treatment divergence is not driven by a pre-existing trend break.

7. Discussion

7.1. Transaction Friction and Adoption Barriers

The net technology effect documented in this paper, approximately 3.1 percentage points of HFCE not attributable to infrastructure subsidies, is consistent with BLIK reducing two distinct categories of transaction cost that previously suppressed digital payment adoption.

The first is informational friction. Traditional card payments in e-commerce require entry of a 16-digit card number, expiry date, and security code, often followed by a 3D-Secure authentication redirect to a third-party page. By replacing this sequence with a six-digit token generated within a familiar mobile banking application, BLIK appears to substantially lower the cost of completing an online purchase. This reduction in checkout friction is consistent with evidence from the Indian context, where digital payment adoption was associated with increased spending frequency, particularly for non-durable goods (Agarwal et al., 2024).

The second is asymmetric trust. In a market with historically elevated consumer concern about sharing financial credentials with unfamiliar online merchants, the bank-led architecture functions as a credible trust intermediary. Moreover, the code used in the BLIK system is valid only for 1 transaction and cannot be reused in contrast to the payment card data. By routing authentication through established banking applications and keeping credentials hidden from merchants, BLIK reduces the perceived security risk of transacting with a wider network of digital retailers. Błach and Klimontowicz (2021) identify this trust architecture as a key determinant of BLIK’s adoption success within the open business model framework.

7.2. Retail Productivity and Channel Shift

Beyond consumer adoption, the mobile standard may have operated as a supply-side productivity shock by accelerating the shift of retail activity toward the digital channel. E-commerce firms typically operate with substantially lower variable costs than brick-and-mortar retailers, reflecting savings on physical infrastructure, inventory management, and cash handling. By providing the dominant payment infrastructure for domestic e-commerce, accounting for approximately 67% of domestic online transaction value by 2024, BLIK likely reduced a key barrier to digital retail expansion. In a competitive market, this reduction in transaction friction lowers the effective marginal cost of digital retail, shifting the aggregate supply curve outward and generating additional transaction volume. This mechanism is consistent with the evidence from Singapore (Agarwal et al., 2019) and India (Agarwal et al., 2025), where mobile payment adoption was associated with measurable increases in business creation and aggregate consumer spending.

7.3. The Pandemic Accelerator

The temporal pattern of the treatment effect is consistent with a technology diffusion interpretation rather than a one-time shock. The divergence between Poland and its synthetic counterpart began in 2015-2016, accelerated during 2018-2019 as network externalities compounded, and widened further during 2020-2022 in association with the COVID-19 pandemic. The contactless payment feature introduced in July 2021, which eliminated the need to enter a six-digit code at the POS terminal, appears to have contributed to a further acceleration of in-store BLIK adoption. The pandemic’s role as a catalyst for cashless payment adoption has been documented across multiple technology and payment domains (Brem et al., 2021; Viardot et al., 2023); Viardot et al. (2023) characterize crises as structural inflection points that compress the technology adoption S-curve by accelerating the transition from early-majority to mass adoption, consistent with the step-up in the treatment gap visible in Figure 2 from 2020 onward. Our two-country comparison isolates the incremental contribution of the sovereign mobile standard above and beyond this general shift. Since donor countries such as Hungary and Latvia faced the same pandemic demand shock but relied primarily on traditional card infrastructure, the synthetic control effectively separates the general pandemic-driven acceleration from the specific effect of the BLIK architecture.

8. Limitations

Our analysis is subject to several limitations inherent to the Synthetic Control methodology and the available data.

First, the outcome variable, i.e. electronic payments as a share of household consumption, is an indirect measure of cash displacement. While an increase in this ratio plausibly reflects a substitution away from cash, it is theoretically possible that BLIK also stimulated aggregate consumption through a reduced "pain of paying" effect (Agarwal et al., 2024). If the payment innovation induced higher total spending rather than pure cash substitution, the observed gap would reflect a combination of market creation and aggregate demand effects. Given the scale of the estimated net effect (approximately 3.1 percentage points of HFCE), a pure demand stimulus of this magnitude is unlikely.

Second, despite our explicit control for the CPSP, other unobserved supply-side confounders may be present. A plausible candidate is the rapid expansion of out-of-home parcel logistics, particularly the InPost parcel locker network, which grew substantially during the treatment period. Since parcel locker usage is correlated with e-commerce activity and BLIK acceptance, our model may attribute some of the logistics-driven e-commerce expansion to the payment technology.

Third, as noted in Section 4, the Polish outcome variable includes both card transactions

and BLIK flows, while donor countries are measured through card payments only. This asymmetry is inherent to the identification strategy, it captures the incremental digitization that BLIK enabled in the absence of any comparable system in the donor pool, but it implies that our estimates constitute an upper bound of the pure net technology effect, since any BLIK flows displacing card transactions would also be counted.

More broadly, the international reference studies to which we compare our results, including evidence from Singapore (Agarwal et al., 2019) and India (Agarwal et al., 2025), are themselves subject to identification challenges. Van Hove (2022) demonstrates that the QR-code payment spillover documented by Agarwal et al. (2019) is methodologically contested, arguing that the observed increase in card sales is better explained by concurrent contactless card adoption rather than by the technology shock itself. This reinforces the value of the quasi-experimental SCM design relative to the event-study and survey-based approaches common in the prior literature, and suggests that our estimates, grounded in a 15-year pre-treatment fit across an international donor pool, rest on stronger causal foundations than the benchmarks typically cited in the payments literature.

Finally, the external validity of our findings is constrained by the institutional specificity of the Polish banking sector. The formation of a six-bank consortium willing to develop and maintain a shared payment standard is rare in markets with more fragmented banking structures. The replication of the BLIK model in other contexts faces coordination and antitrust considerations that may not be present in Poland’s concentrated banking sector.

9. Conclusion

This paper used the introduction of BLIK in Poland as a natural experiment to estimate the causal effect of a sovereign mobile payment standard on transaction digitization. Using a penalized synthetic control method with a donor pool of 21 European economies, we document a progressive divergence between Poland and its synthetic counterpart that begins in 2015 and continues through 2024. Our two-stage identification strategy decomposes the aggregate gap: the Cashless Payment Support Program’s terminal subsidies account for approximately 2.7 percentage points (approximately 47% of the 2023 gap), while the mobile payment technology itself is associated with the remaining 3.1 percentage points (approximately 53%). This net technology effect is consistent with genuine market creation, the digitization of transactions that would likely have remained in cash in the absence of the innovation.

Our findings carry implications along three dimensions.

Substitution versus market creation. The decomposition is inconsistent with a pure substitution hypothesis. A net expansion of approximately 3.1 percentage points of HFCE in digital transaction volume, confirmed across multiple placebo and sensitivity tests, suggests that low-friction mobile payment standards do not merely displace existing card

transactions; they appear to extend digital commerce into segments where cards were previously impractical, including micro-merchants, informal service providers, and peer-to-peer transfers. This pattern is consistent with the technology leapfrogging literature and with evidence from comparable systems in Brazil and India (Reserve Bank of India, 2024; Sarkisyan, 2025).

Payment governance models. As the European Central Bank progresses toward a digital euro (European Central Bank, 2023b) and the European Payments Initiative scales the Wero wallet (European Payments Initiative, 2025), policymakers face a choice among three governance architectures: Big Tech overlays (Apple/Google Pay), state utility models (Pix/UPI), and bank-consortium standards (BLIK, Bizum, Vipps MobilePay). The Polish evidence is consistent with the view that the consortium model can achieve financial digitization outcomes comparable to state-run utilities while preserving commercial sustainability. BLIK’s progression from peer-to-peer transfers to e-commerce dominance represents a diffusion trajectory that European payment policy may seek to replicate at continental scale.

A related policy question concerns the potential digitization effects of central bank digital currencies (CBDCs). The digital euro, currently advancing through its preparation phase (European Central Bank, 2023b), may produce analogous market-creation dynamics, though with important contextual qualifications. In economies that lack an established domestic mobile payment alternative, where the incumbent digital options remain card-based or absent, a widely adopted CBDC could replicate the pattern documented here for BLIK: converting cash-only transactions into digital flows and extending digital commerce into segments that existing card infrastructure has not reached. However, even in economies where domestic mobile payment schemes are already operational, the introduction of a CBDC is unlikely to be neutral. The evidence documented in this paper suggests two concurrent effects are plausible: additional substitution, as a state-backed currency standard displaces a share of incumbent mobile payment volume, and further market creation, as the CBDC reaches transaction segments, such as smaller merchants, informal services, older demographics, where existing schemes have not yet achieved critical mass. Whether the net effect of CBDC adoption resembles the market-creation dynamics of BLIK or reduces primarily to inter-scheme substitution is an empirical question whose resolution will depend critically on the institutional and infrastructural context of the adopting economy. Disentangling these two channels represents a first-order research agenda for the period following CBDC launch.

Forecasting and diffusion dynamics. BLIK’s growth from zero to 18.5 million active users and approximately 67% of domestic e-commerce value within a decade illustrates the non-linear dynamics predicted by the Bass diffusion model and the scaling law documented by Van Hove et al. (2026). Once a sovereign payment standard crosses its network tipping point, the compound effect of cross-side externalities generates an accelerating adoption

curve. The empirical diffusion trajectory documented here—in particular, the decade-long acceleration from market launch to dominance—provides a quantitative benchmark for forecasting the adoption curves of nascent sovereign payment systems, including the European Payments Initiative’s Wero wallet (European Payments Initiative, 2025) and the prospective digital euro (European Central Bank, 2023b). As BLIK expands internationally and as comparable systems mature across Europe, the Polish case offers a validated reference point for policymakers and technology forecasters estimating the pace and scale of digital payment diffusion under interoperable, bank-led governance.

Declaration of Interest

This research builds on analytical work conducted by members of the EY Economic Analysis Team for a report commissioned by Polski Standard Płatności (PSP), the operator of the BLIK standard. The econometric methodology and empirical analysis presented in this paper represent independent academic work conducted by the authors. PSP had no influence on the research design, the interpretation of results, or the decision to submit this manuscript for publication.

Replication materials

The dataset and STATA codes that allow replication of the results are available at <http://doi.org/10.17632/mhhdbj92jt.1>

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